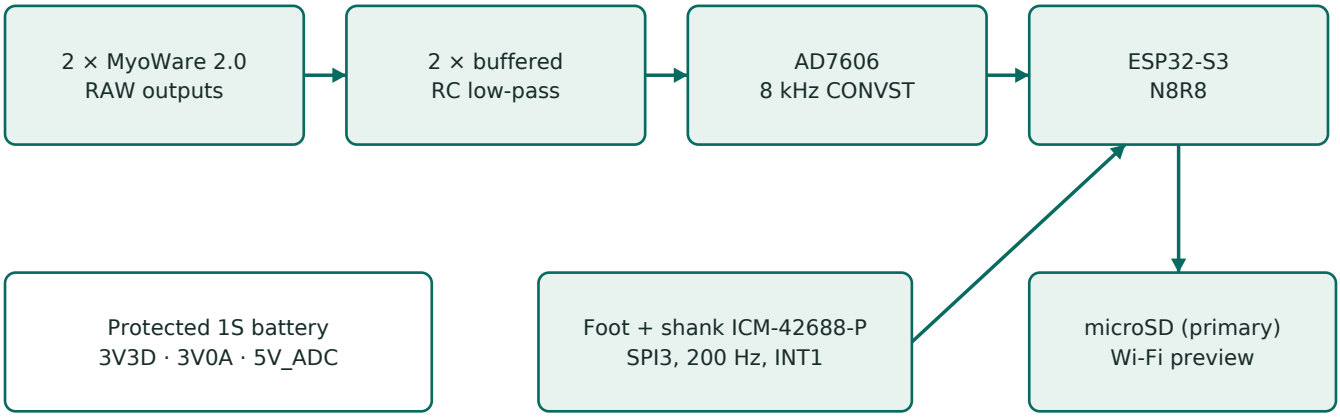
AFO RESEARCH RECORDER

Main setup reference

AD7606 · two MyoWare 2.0 EMG channels · two ICM-42688-P motion sensors

Package `ad7606-two-channel-icm42688` · firmware `ad7606-2ch-1.12` · 12 October 2026

Project lead: Muttakin Rahman



Functional paths only; not physical header positions.

> **Engineering prototype.** Firmware, file format, host tools, PCB rules, wiring and simulations have automated evidence. No assembled recorder has yet passed the physical bench sequence. Nothing here claims measured signal quality, battery runtime, wearable safety or clinical performance.

This document collects, in one place, what the main setup is, how it is wired and powered, which firmware image to flash at each build stage, what the simulations have shown and what is still open. It is generated from the package files, so its tables match the repository at the date above.

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1. The setup at a glance

This is the reference configuration of the project: the fixed two-input recorder. Two bare MyoWare 2.0 sensors feed their RAW outputs through two buffered low-pass stages into an AD7606 simultaneous-sampling ADC. An ESP32-S3 triggers conversions at 8 kHz, reads two ICM-42688-P carriers (foot and shank) at 200 Hz each, writes every record to microSD and sends a best-effort live copy to a laptop over Wi-Fi.

Item	Value
EMG inputs	2 fixed (EMG1, EMG2), single-ended MyoWare RAW; four-channel builds are rejected
EMG sampling	8 kHz per channel; both channels converted simultaneously on one CONVST edge
ADC	AD7606, 16-bit, ±5 V range, oversampling off; 152.588 µV per code; serial mode 2 at 8 MHz
Analog path	Two 3.3 kΩ / 47 nF poles with unity buffers; about −1.85 dB at 500 Hz, −3 dB near 660 Hz
Motion	Two ICM-42688-P (MIKROE-4237 carriers), 200 Hz accel + gyro, ±16 g / ±2000 °/s, SPI3
Controller	ESP32-S3-WROOM-1 N8R8 (PCB) or ESP32-S3-DevKitC-1 N8R8 (breadboard)
Primary storage	microSD over SDMMC 1-bit; 64-byte CRC-protected records; 2.44 s PSRAM record queue
Live preview	Private WPA2 access point `AF0-Recorder-B`; per-device password generated on first boot
Power	Protected 1S Li-ion; 3V3D digital, 3V0A analog, 5V_ADC for AVCC; USB is sense-only
Firmware	`ad7606-2ch-1.12`, ESP-IDF v5.4.2, nine profiles
Simulation	All 18 integrated Wokwi cases pass on the 1.12 image (12 October 2026)
Physical status	Not built or measured; compact PCB not ordered

2. Parts

Laboratory (breadboard) build. These are the selected modules for the staged bench build. Use one package end to end; do not mix MPU-6050 or four-channel parts or firmware.

Function	Selected part	Boundary
Controller	ESP32-S3-DevKitC-1 N8R8	Use matching breadboard firmware; SD CMD is GPIO47
Protected battery	Adafruit 328, 1S 2500 mAh	Check connector polarity; external charging only
5 V regulator	Pololu 4082 S13V30F5	Lab power implementation, not PCB converter validation
3.3 V regulator	Pololu 4980 S13V25F3	Do not parallel with DevKit 3.3 V output
Analog LDO	TPS7A2030PDBVR on Adafruit 1230	Specified pad/header mapping and local soldered bypass
SD	Adafruit 4682	3.3 V interface; SDIO wiring in current tables
USB inlet	Adafruit 1833	External inlet, VBUS sense-only; both DevKit sockets unused
EMG	Bare MyoWare 2.0	VIN/GND/RAW pads; ENV/RECT unused
Breadboards	BusBoard BB830	63 rows; unused long power rails
ADC	RoboticsBD RBD-3184 / HW-AD7606-F4	Photo-labeled headers; serial configuration and VIO measured before connection
IMUs	Two MIKROE-4237 6DOF IMU 14 Click modules	Manufacturer SPI configuration and mikroBUS header map

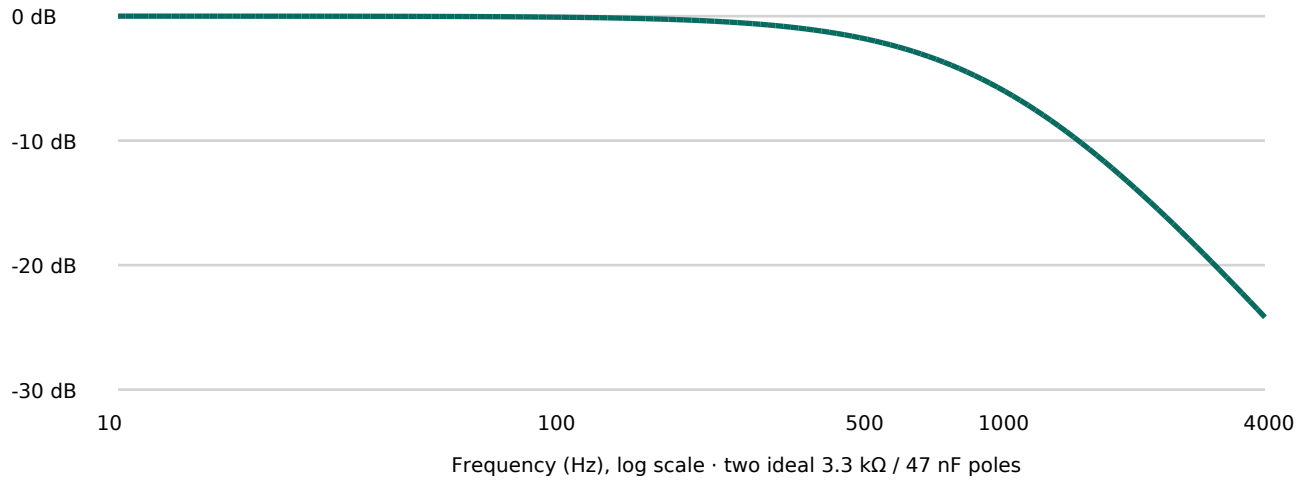
Compact PCB, key parts (full list: `hardware/main/bom.csv`, 108 lines). D30 and R70 were added on 12 October 2026 for the AD7606 supply-ramp fix.

Ref	Value	Part number
U1	ESP32-S3-WROOM-1-N8R8	ESP32-S3-WROOM-1-N8R8
J2	microSD	DM3AT-SF-PEJM5
J1	USB-C (battery powered)	USB4105-GF-A
Q2	2N7002	2N7002,215
D1	USB ESD	USBLC6-2SC6
J3	FOOT IMU	BM08B-GHS-TBT(LF)(SN)
J4	SHANK IMU	BM08B-GHS-TBT(LF)(SN)
U2	AD7606BSTZ	AD7606BSTZ
U5	MCP6001T-I/OT	MCP6001T-I/OT
U3	MCP6004T-I/ST	MCP6004T-I/ST
U4	MCP6004T-I/ST	MCP6004T-I/ST
U6	TPS63070RNMR	TPS63070RNMR
U7	TPS7A2030PDBVR	TPS7A2030PDBVR
U8	TPS60150DRVR	TPS60150DRVR
R70	100k	RC0603FR-07100KL
D30	BAT760	BAT760,115

3. EMG signal chain

Each MyoWare RAW output passes a 3.3 kΩ / 47 nF low-pass section, a unity buffer, a second 3.3 kΩ / 47 nF section and another unity buffer, then 100 Ω in series and 1 nF to ground at the AD7606 input. V1 and V2 are single-ended; V3–V8 are grounded on this board. All eight conversion words are clocked out, and only V1 and V2 are exported.

A 1 MΩ resistor biases an unplugged RAW port toward the buffered 1.5 V midpoint. This is not an offset stage: the converter reconstructs the connector voltage with `input_scale=1` and `input_midpoint_mv=0`, and records the bias separately as `unplugged_input_bias_mv=1500`.



Frequency	Nominal gain	Expected rms_ac_codes (100 mVpp sine)
20 Hz	−0.00 dB	231.6
100 Hz	−0.08 dB	229.5
500 Hz	−1.85 dB	187.2

Frequency	Nominal gain	Expected rms_ac_codes (100 mVpp sine)
1000 Hz	−5.80 dB	118.8

Scaling. Volts at the ADC input = code × 5 / 32768 (152.588 µV per code). A 0–3 V sensor uses about 30 % of the bipolar code range, so “16-bit” describes the word, not 16 effective bits at the muscle. Reference, gain, offset and noise need bench calibration.

Protection. BAV199 clamps (ground → junction → 3V0A) sit at the first filter node. They do not make the connector tolerant of arbitrary ±5 V or ±10 V signals; keep test signals within 0–3 V.

4. Power and the AD7606 supply ramp

- A protected single-cell pack enters through a reverse-polarity P-channel MOSFET. Charging is external.
- TPS63070 buck-boost makes 3V3D (nominal 3.328 V) for the ESP32, IMUs, SD and AD7606 VDRIVE.
- TPS7A2030 LDO makes 3V0A for the MyoWare sensors and buffers; the buffered midpoint is 1.5 V.
- TPS60150 charge pump makes 5V_ADC (AVCC, 4.75–5.25 V) from 3V3D.
- USB VBUS is sense-only (2N7002, active-low GPIO21). Recording is refused while USB is attached.

Supply-ramp fix (PCB, 12 October 2026)

Because AVCC is generated from 3V3D, VDRIVE rises before AVCC, which conflicts with the AD7606 absolute maximum VDRIVE ≤ AVCC + 0.3 V and with ADI AN-1091. The PCB now holds the converter in full shutdown until both rails are up:

- AD7606 STBY (pin 7) runs on net ADC_STBY to GPIO14, with R70 (100 kΩ) pulling it low; RANGE stays low.
- D30 (BAT760 Schottky) from 3V3D to 5V_ADC limits VDRIVE − AVCC while the charge pump starts.
- Firmware (PCB builds, from 1.11) releases STBY once per boot, waits the 30 ms wake-up time, then resets the ADC.
- KiCad DRC 0 violations / 0 unconnected, ERC 0. Breadboard builds are unchanged: the module straps STBY.

Bench rail	Stage-1 acceptance
3V3D digital	3.23–3.43 V at 0 / 100 / 300 mA; ripple ≤ 50 mVpp (20 MHz bandwidth)
3V0A analog	2.94–3.06 V at 0 / 20 / 50 mA; midpoint 0.49–0.51 × rail; ripple ≤ 10 mVpp
5 V ADC	4.8–5.2 V at 10–50 mA; never above 5.25 V during start-up

5. ESP32-S3 pin map

From `design/gpio-map.json` (compact PCB). The only breadboard difference is SD CMD: **GPIO47 on the DevKit breadboard, GPIO38 on the PCB**, because the DevKit drives an RGB LED on GPIO38. Flash the matching build.

GPIO	Net	GPIO	Net
0	BOOT	16	FOOT_INT
1	BAT_SENSE	17	SHANK_INT
2	LED_LOW	18	BUTTON
4	IMU_MOSI	19	USB_DM_MCU
5	IMU_MISO	20	USB_DP_MCU

GPIO	Net	GPIO	Net
6	IMU_SCK	21	USB_PRESENT_N
7	FOOT_CS	38	SD_CMD
8	ADC_DRDY	39	SD_CLK_MCU
9	ADC_RESET	40	SD_D0
10	ADC_CS	41	LED_REC
11	ADC_CONVST	42	LED_ERR
12	ADC_SCK	43	UART_TX
13	ADC_MISO	44	UART_RX
14	ADC_STBY	47	EXP_SDA
15	SHANK_CS	48	EXP_SCL

AD7606 CVA and CVB share GPIO11 (CONVST). The net named ADC_DRDY carries BUSY: the AD7606 has no separate data-ready output. SPI2 runs at 8 MHz, mode 2 (idle high, falling-edge sample). The IMUs share SPI3 on GPIO4/5/6 with separate chip selects (foot GPIO7, shank GPIO15) and INT1 lines (foot GPIO16, shank GPIO17). Full contact-level tables: `docs/build-guide.html`.

6. Motion sensors and timing

- MIKROE-4237 carriers with JP2–JP4 in the SPI position (pads 1–2), 3.3 V supply, mikroBUS orientation as drawn.
- Firmware checks WHO_AM_I = 0x47 and configuration readback, then sets 200 Hz, ±16 g, ±2000 °/s, 16-byte FIFO packets.
- FIFO timestamp ticks last 32/30 µs on the ICM-42688 (TDK driver PLL scale; datasheet section 12.7). Expect a tick step of about 4,687.5 per 200 Hz sample.
- The datasheet allows ±1.5 % clock error at 25 °C and ±2 % over temperature; the stage-6/7 gates use ±2 %.
- Each sensor has its own clock. The converter adds `host_time_fitted_us`, a least-squares fit of every interrupt anchor, to align IMU samples with EMG. Times remain flagged as estimates until bench timing is characterised.

7. Firmware 1.12 and its profiles

All profiles are built from one source set, `ad7606-2ch-1.12`, with ESP-IDF v5.4.2 for the ESP32-S3. Flash the bootloader, partition table and application from **one** profile at its manifest offsets (`0x0`, `0x8000`, `0x10000`). Keep the application hash with every measured run. The binaries are cross-compiled only; none has been flashed to an assembled recorder.

Stage	Profile	Use	application.bin SHA-256 (first 16)
2	`diagnostic-controller`	Controller, button, LEDs, USB sense	`97c3cc29ae92b704`
3–5	`diagnostic-adc`	AD7606 and the analog paths, added one at a time	`4ae7661262376a3b`
6	`diagnostic-imu-one`	Adds the foot ICM carrier	`b70a485bef6f2278`
6	`diagnostic-imu-two`	Adds the shank ICM carrier	`ae43b54b0baf6006`
any	`diagnostic-imu-bench-one`	DevKit + one carrier only; 60 s PASS/FAIL verdicts	`db229b898f14e29f`

Stage	Profile	Use	application.bin SHA-256 (first 16)
any	`diagnostic-imu-bench-two`	DevKit + both carriers only; 60 s PASS/FAIL verdicts	`1c4ea59ca9392751`
7	`breadboard-sd-only-2ch`	Full breadboard, SD recording, Wi-Fi off	`8d71a8a85816d5eb`
8–10	`breadboard-record-2ch`	Full breadboard, SD + best-effort Wi-Fi preview	`90b66ff2bd1a2efd`
PCB	`record-2ch`	Compact PCB only (SD CMD on GPIO38)	`ed3dc90453ab2067`

What changed recently

Version	Change
1.12	The Wi-Fi live task announces a new session (metadata packet) before that session's first data packet. On 1.11 a record dequeued just after the metadata check could go first, and the laptop dropped it as an unknown session. Found by the Wokwi IMU missing-interrupt case.
1.11	PCB builds release AD7606 STBY on GPIO14, wait 30 ms, then reset (ADI AN-1091).
1.10	Official ICM-42688-P datasheet cross-check; clock gates widened to ±2 %; `within_datasheet_tolerance`.
1.9	IMU-only bench diagnostics with PASS/FAIL verdicts; register-level ICM model.
1.8	ICM FIFO ticks converted as 32/30 µs instead of 1 µs.
1.7	PSRAM record queue of 20,480 records (2.44 s); no compiled Wi-Fi password; 1,400-byte metadata limit.

Rebuilding

```
cd firmware
idf.py -B build-breadboard-2 -DEMG_CHANNEL_COUNT=2 -DAFO_HARDWARE=breadboard \
-DAFO_WIFI=1 -DAFO_DIAGNOSTIC_STAGE=0 -DAFO_IMU_COUNT=2 build
idf.py -B build-pcb-2 -DEMG_CHANNEL_COUNT=2 -DAFO_HARDWARE=pcb \
-DAFO_WIFI=1 -DAFO_DIAGNOSTIC_STAGE=0 -DAFO_IMU_COUNT=2 build
```

`AFO_DIAGNOSTIC_STAGE` selects 2 (controller), 3 (ADC), 4/5 (analog) or 6 (IMU); diagnostics do not mount SD or start Wi-Fi. Use a separate build directory for each combination.

8. Recording, live preview and data

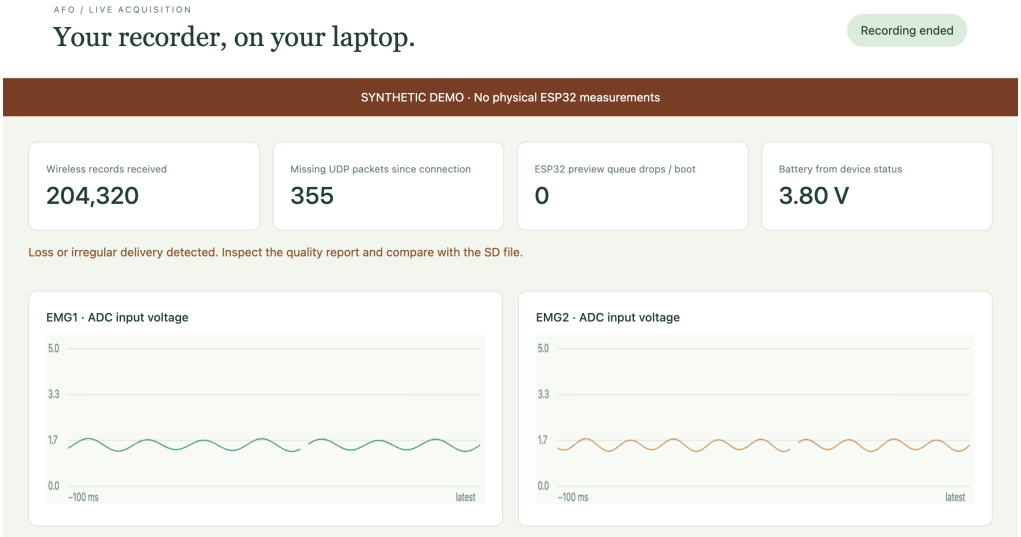
- The button starts and stops a session. Recording is refused while USB is attached or the battery is low.
- Files keep the AFOLOG1 header and 64-byte CRC-protected records (schema 3, record kind 7 for AD7606 frames). Each EMG record carries both codes, read duration, elapsed slots, channel mask 0x03 and a read-start timestamp.
- The SD file is the primary record. Wi-Fi is a second, best-effort path: preview losses are shown and never alter the SD stream.
- Wait for the recording LED to go out before removing power or the card. An error LED, a nonzero stop reason or a missing END needs investigation.
- At first boot the recorder prints its Wi-Fi password on the USB console (`Laptop AP: AFO-Recorder-B; password: …`). Erasing flash creates a new one.

Converting a recording

```
python3 host/convert.py trial.afolog --out trial-csv --no-plots
```

Output: `emg.csv`, `foot.csv`, `shank.csv`, `status.csv`, `metadata.json` and `quality.json`. Exit code 0 means the loss and signal checks passed; 2 means the output has reported quality problems; 1 means conversion failed. EMG values are volts at the RAW connector, not input-referred muscle microvolts.

Live view: with USB removed, join `AF0-Recorder-B` and run `python3 host/live_receiver.py --device 192.168.4.1 --out trial-wifi`; the live page opens on port 8766.



Laptop live preview (software replay, not a measurement).

9. Build sequence

Build one stage at a time with power off for each addition, and record the result before continuing. An empty results cell means untested. No person is attached during any instrument-connected test.

Stage	Goal	Firmware	Pass when
0	Identify the exact build	None	Inventory only; no power. Mark every inventory.csv row acquired or not acquired; for missing items keep the proposed part number. Photograph both sides of every module, connector and jumper with readable markings. Identify the protected battery and charger separately. List instruments, calibration dates and available dummy loads.
1	Qualify power one rail at a time	None	• No rail short; correct source polarity; lab supply only, battery disconnected • 3.23–3.43 V at 0/100/300 mA; ripple ≤50 mVpp (20 MHz bandwidth); provisional project screen • 2.94–3.06 V at 0/20/50 mA; midpoint 0.49–0.51 × measured analog rail (1.47–1.53 V at a nominal 3.00 V rail); ripple ≤10 mVpp; provisional project screen (+1 more checks in the workbook)
2	Controller and controls	diagnostic-controller	• 20 programming/reset cycles without failure; buttons and LEDs match controls • GPIO21 LOW attached / HIGH absent; attachment indicator 1/0; external usb-pu 10k GPIO21–3V3D measured unpowered (the diagnostic enables an internal pull-up, recording firmware does not); no USB VBUS feed into regulated 5 V; DevKit USB sockets empty
3	ADC module and two DC inputs	diagnostic-adc	• AVCC 4.75–5.25 V; VIO 3.23–3.43 V (project target); internal reference and serial strap verified; ESP32-safe output levels; RANGE/OS low • GPIO11 through DIST1 a29/b29/c29 reaches both CVA/CVB; common rising edge • 1 V / 2 V ideal 6554 / 13107; compute from measured source; ±65 codes plus measured uncertainty; swapping sources swaps EMG identities (+4 more checks in the workbook)
4	First analog input	diagnostic-adc	• 1.5 V + 100 mVpp synthetic input at 20/100/500/1000 Hz via ADC (4 kHz on scope only); unity DC; measured response within simulated tolerance envelope plus ±0.5 dB margin; no person attached • Unplugged RAW settles 1.50 V ±30 mV; use safe 0.1–2.9 V sources for connected 3 V analog stage

Stage	Goal	Firmware	Pass when
5	Second analog input and interaction	diagnostic-adc	• Repeat BB1 tests on BB2; exactly EMG1/2; common frame timestamp; no extra exported muscle channels • Provisional <−40 dB crosstalk screen; generator feedthrough measured separately; preserve RMS noise and 10 min error log
6	Foot IMU, then shank IMU	diagnostic-imu-one, then diagnostic-imu-two	• SPI JP2–JP4 1–2; 3.3 V; WHO_AM_I 0x47/readback; foot CS GPIO7 / INT1 GPIO16; correct axis signs • Shank CS GPIO15 / INT1 GPIO17; 200 Hz ±2% per sensor over 60 s (datasheet internal clock ±1.5% at 25 °C, ±2% over temperature) (packets ÷ interval_us, or INT1 capture); tick_step_mean 4687.5 ±0.5% and mean host_us_per_tick 1.067 ±2% (32/30 µs ticks; if both read 5000 and 1.00 instead, report it: the firmware tick constant must change) per sensor over ≥60 windows; no FIFO/missing-IRQ/invalid-packet errors • ±16 g / ±2000 °/s configured; gravity axis ±1 g within ±0.1 g, cross axes within ±0.1 g; gyro per axis <5 °/s before calibration
7	Original SD recording	breadboard-sd-only-2ch	• AD7606 / two inputs / ICM / breadboard metadata; RAW code ×5/32768 V; scale 1 midpoint offset 0; retain raw IMU counters IRQ anchors read start/end and timing flags • CMD GPIO47 CLK GPIO39 D0 GPIO40; one 60 min run and one full 2 h run ending with stop reason 5 (session limit); container CRC32 and counters pass; no unexplained saved gaps/queue errors; final queue_high_water ≤10,240 (half of 20,480); END and expected stop reason; no sync/close failure; card make/model/capacity/speed class/filesystem recorded • Refuse start with USB attached; attach during synthetic recording gives explicit stop/error; no person attached (+2 more checks in the workbook)
8	Wi-Fi while SD records	breadboard-record-2ch	• 60 min; only EMG1/2; explicit preview losses; independently audit original SD counters/CRC/END/stop and confirm no unexplained saved loss • Disconnect/reconnect produces visible preview gaps; compare identities codes timestamps and stop/error indications without filling missing data
9	Battery and full-load regression	breadboard-record-2ch	• Both lab-supply leads removed; ≥2 h measured runtime; 20 repeatable startups/shutdowns; complete 60 min trial; record pack voltage age temperature and events
10	MyoWare integration and research readiness	breadboard-record-2ch	• Correct VIN/GND/RAW pads and supply; no saturation or cabling fault; distinguish RAW volts from electrode-referred EMG • Consent and safety review complete; mounting strain relief/AFO clearance checked; no lab instruments mains USB or charging while electrodes are attached

10. Verification evidence

Wokwi integrated matrix on firmware ad7606-2ch-1.12: 18 of 18 cases pass (Wokwi web editor, 12 October 2026). Each case runs the production recorder, sensor drivers, serializer and Wi-Fi task on an instrumented image; each serial export is checked with the production converter and live decoder, and each case keeps its logic-analyser trace.

Case	Result	Outcome
Nominal recording	Pass	10,039 EMG / 262 foot / 251 shank saved; normal stop
50 ms SD write delay	Pass	10,038 EMG / 262 foot / 251 shank saved; normal stop
350 ms queue stall	Pass	10,039 EMG / 262 foot / 251 shank saved; normal stop
SD write error	Pass	111 EMG / 14 foot / 3 shank saved; stopped with reason storage, as expected
USB attached mid-recording	Pass	3,187 EMG / 91 foot / 80 shank saved; stopped with reason usb, as expected
Battery low	Pass	24,004 EMG / 612 foot / 601 shank saved; stopped with reason battery, as expected
ADC BUSY stuck high	Pass	0 EMG / 11 foot / 0 shank saved; stopped with reason timing, as expected

Case	Result	Outcome
IMU FIFO overflow	Pass	0 EMG / 0 foot / 0 shank saved; stopped with reason sensor, as expected
IMU interrupt missing	Pass	805 EMG / 0 foot / 20 shank saved; stopped with reason sensor, as expected
No laptop listening	Pass	10,040 EMG / 262 foot / 251 shank saved; normal stop
Wi-Fi packet loss	Pass	10,039 EMG / 262 foot / 251 shank saved; normal stop
Power cut mid-recording	Pass	7,984 EMG / 211 foot / 200 shank saved; no END; synced records recovered, as expected
SD mount failure	Pass	Initialization failed; recording disabled. (expected)
Full card at start	Pass	Start refused (expected)
Filtered synthetic EMG	Pass	10,039 EMG / 262 foot / 251 shank saved; normal stop
Shank IMU absent	Pass	Initialization failed; recording disabled. (expected)
ADC BUSY never rises	Pass	0 EMG / 11 foot / 0 shank saved; stopped with reason timing, as expected
ADC ignores CONVST	Pass	0 EMG / 11 foot / 0 shank saved; stopped with reason timing, as expected

Other checks

Check	Result
KiCad ERC / DRC	0 / 0 violations; no unconnected items
Schematic vs PCB pads	395 pins compared, 0 mismatches; firmware GPIO map matches
Host tools	48 converter, live-decoder, legacy and firmware-limit tests pass (rerun 12 October 2026)
Native recorder control	38 production-code cases (19 per board) pass
Native drivers	17 AD7606 driver groups and 9 IMU timing groups pass (ASan/UBSan)
ICM register model	Production driver passes with no datasheet-rule violation; fault cases detected
60 s native pipeline	480,000 EMG frames, 12,009 foot, 12,001 shank; no unexplained loss
ngspice analog	Both channels, 16 tolerance corners, 54 DC/loading cases
Physical tests	None performed

These scopes are distinct and none is a measurement of an assembled recorder. Earlier failed and partial runs (1.11, v1.5) are kept in the repository rather than relabelled.

11. Open items before use

- Qualify the received AD7606 module electrically (serial strap, VIO routing, reference) before connecting it.
- Measure every rail, start-up and ripple (stage 1), including the 5 V ADC rail never exceeding 5.25 V.
- Measure actual 8 kHz trigger timing and jitter under SD and Wi-Fi load; the 125 µs budget is only arithmetic so far.
- Measure analog noise, clipping and crosstalk; set study-specific limits before claiming research suitability.
- Characterise each IMU clock (200 Hz and µs-per-tick within ±2 %) and EMG-to-IMU alignment.
- Run a 60-minute SD endurance recording and the full-card and power-cut recovery checks.
- Measure battery runtime (target at least two hours) and 20 repeatable start/stop cycles.
- Complete the final PCB layout review before ordering, then repeat power, ripple, antenna, thermal and noise checks on the built board.

- Obtain institutional review before any wearable or participant use. Engineering stage completion does not authorise participants.

12. Where things are

Path (inside designs/ad7606-two-channel-icm42688/)	Contents
`development/index.html`	Stage-by-stage workbook with blank results sheets
`viewer/build.html`	Interactive assembly workspace (serve over HTTP)
`docs/build-guide.html`	Complete hole/contact connection tables
`docs/lab-quickstart.html`	Short lab sequence with pass limits and diagnostic field meanings
`docs/circuit-audit.html`	Circuit review, datasheet cross-checks and the supply-ramp change
`docs/integrated-verification.html`	Wokwi matrix report and history
`docs/understanding-the-recorder.pdf`	Longer supervisor handbook (background and study design)
`docs/circuit-diagrams.pdf`	Functional map and full KiCad schematic
`firmware/builds/`	Nine profiles with manifests and hashes
`hardware/, manufacturing/`	KiCad sources; Gerber, drill, BOM, placement and STEP exports
`host/`	Converter, live receiver, plots and tests
`breadboard/breadboard-lab-kit.zip`	Self-contained lab kit for the breadboard stages

Live project: https://shadowfax-505.github.io/afo-recorder/ · Source: https://github.com/shadowfax-505/afo-recorder